S&P 500 Return Prediction

<u>Data Description</u> (1977.2.1 ~ 2026.6.30)

1) S&P500 Return

2) Inflation Rate
○ Monthly(annual) inflation rate(12 month moving annual rate). Customer Price Index from the Bureau of Labor Statistics. Connor(1995), Welch &
Goyal(2008) investigated the explanatory power of the inflation rate.
○ Data Source: US Inflation Calculator
CPI is missing for 2025.10, we fill the number by linear interpolation

3) Unemployment Rate
○ Monthly(annual) US unemployment rate. Connor(1995) investigated the explanatory power of the unemployment rate.
○ Data Source: FRED, Federal Reserve Bank of St. Louis

4) Price/Earnings Ratio of S&P500
○ Log of price minus Log of earning, where Earning is the 12-month moving sum. Welch & Goyal(2008) investigated the explanatory power of the earning price ratio.
○ Data Source: Earning from Robert Shiller's Website; Price from Yahoo Finance

5) Dividend Yield of S&P500
○ Log of dividend minus log of lagged price, where Dividend is the 12-month moving sum. Connor(1995), Welch & Goyal(2008) investigated the explanatory power of dividend yield.
○ Data Source: Dividend from Robert Shiller's Website; Price from Yahoo Finance

6) Risk Free Rate(3-month Treasury Bill Rate)
○ Monthly(annual) *3-Month Treasury Bill: Secondary Market Rate (TB3MS).* Welch & Goyal(2008) investigated the explanatory power of the Treasury Bill. ○ Data Source: FRED, Federal Reserve Bank of St. Louis.

7) Dividend Price Ratio of S&P500
○ Log of dividend minus log of price, where Dividend is the 12-month moving sum. Welch & Goyal(2008) investigated the explanatory power of the Dividend Price Ratio.
○ Data Source: Dividend from Robert Shiller's Website; Price from Yahoo Finance

8) Dividend Payout Ratio of S&P500
○ Log of dividend minus log of earning, where Dividend & Earning are the 12-month moving sum. Welch & Goyal(2008) investigated the explanatory power of the Dividend Payout Ratio.
○ Data Source: Robert Shiller's Website

9) Long-term Government Bond Yield
○ Monthly(annual) 30-year US Treasury Bill Yield. Linear interpolation would be used for the null value. Welch & Goyal(2008) investigated the explanatory power of the Long-term Yield.

○ Data Source: FRED, Federal Reserve Bank of St. Louis.

10) Spread Between Long-term Government Bond Yield and 3 Month Treasury Bond Yield
○ Monthly(annual) 30-year US Treasury Bill Yield minus monthly(annual) 3 month Treasury Bond Yield. Linear interpolation would be used for the null value. Welch & Goyal(2008) investigated the explanatory power of the Term Spread.
○ Data Source: 3 months & 30 year Treasury Bond yield from FRED, Federal Reserve Bank of St. Louis

11) S&P500 Stock Variance
○ Sum of the square of daily stock return(in the past 1 year). Welch & Goyal(2008) investigated the explanatory power of the Stock Variance.
○ Data Source: daily price from Yahoo Finance

12) Long Term Rate of Return
○ Monthly(annual) Return on long-term yield. Linear interpolation would be used for the null value. Welch & Goyal(2008) investigated the explanatory power of the Long Term Rate of Returns.
○ Data Source: 30 year Treasury Bond yield from FRED, Federal Reserve Bank of St. Louis

Reference

1. Inflation Rate
US Inflation Calculator.(n.d.). *Historical Inflation Rates: 1914-2020.* https://www.usinflationcalculator.com/inflation/historical-inflation-rates/

2. Unemployment Rate
FRED, Federal Reserve Bank of St. Louis.(n.d.). *Unemployment Rate [UNRATE].* https://fred.stlouisfed.org/series/UNRATE

3. S&P 500 Price/Earning Ratio
Earning:
Robert Shiller's website(n.d.). *ONLINE DATA ROBERT SHILLER.* http://www.econ.yale.edu/~shiller/data.htm
Price: Yahoo Finance

4. S&P 500 Dividend Yield
Dividend:
Robert Shiller's website(n.d.). *ONLINE DATA ROBERT SHILLER.* http://www.econ.yale.edu/~shiller/data.htm
Price: Yahoo Finance

5. Risk Free Rate(3 Month T-bill Rate)
FRED, Federal Reserve Bank of St. Louis.(n.d.). *3-Month Treasury Bill: Secondary Market Rate (TB3MS).* https://fred.stlouisfed.org/series/TB3MS

6. Dividend/Price Ratio
Dividend:
Robert Shiller's website(n.d.). *ONLINE DATA ROBERT SHILLER.*

http://www.econ.yale.edu/~shiller/data.htm
Price: Yahoo Finance

7. Dividend Payout Ratio(Dividend/Earning)
Robert Shiller's website(n.d.). *ONLINE DATA ROBERT SHILLER.*
http://www.econ.yale.edu/~shiller/data.htm

8. 30 years T-bill Rate
FRED, Federal Reserve Bank of St. Louis.(n.d.). *30-Year Treasury Constant Maturity Rate (GS30).*
https://fred.stlouisfed.org/series/GS30
#The U.S Treasury suspended issuance of the 30 year bond between 2/15/2002 and 2/9/2006.

9. Spread between 30 years T-bill Rate & 3 Month T-bill Rate
3 month & 30 year Treasury Bond yield: FRED, Federal Reserve Bank of St. Louis